



ABSENS ATOMUS

Paper II

Boundary Genesis

Representation Individuates, and Why That Puts Absolute Absence Out of Reach

An attempt to turn a repeated failure into a theorem. Written to be attacked at the principle.

What this paper is trying to do, and what it isn't

There is a companion experiment to this paper. Four different systems were asked, independently and without seeing one another, to imagine or represent absolute absence. All four made several genuine attempts by different routes. All four failed. More importantly, all four failed at the same place and in the same way: the boundary that spoiled each attempt was produced by the act of reaching, not encountered as resistance from the target. And all four found, on their own, that even a hypothetical success could not be confirmed, because confirmation is itself an act of individuation.

That is strong evidence, but it is evidence by exhaustion. It shows that every representer who tried, failed the same way. It does not show that every possible representer must. This paper is the attempt to close that gap, to move from every mind that tried, failed to any representer must fail. I want to be honest at the outset that whether it fully closes is exactly what a reader should test. The proof below is only as strong as one principle, and I will mark that principle clearly rather than hide it.

1. Primitives and Definitions

I will keep the machinery as small as I can. The whole argument runs on a handful of definitions and one substantive principle.

Definition 1 (System). A system S is any collection of elements together with the relations that hold among them. Languages, logics, categories, models, minds considered as bearers of representations, and formal theories are all systems in this sense. Nothing here assumes the system is finite, symbolic, or human.

Definition 2 (Individuation). An element x of a system S is individuated in S if its identity within S is fixed by its distinctions from other elements, that is, by the relations of the form x differs from y for y in S . To be individuated is to be settled as this element rather than the others.

Definition 3 (Boundary). The boundary of an element x in S is the totality of distinctions that individuate it, the collection of all facts of the form x differs from y that hold in S . Boundary here is not spatial. It is relational. To have a boundary is to stand in at least one distinction.

Definition 4 (Representation). A representation of a target T in a system S is a correspondence r that places T as some element $r(T)$ of S . To represent T is to make T available as an element that S can hold, refer to, or operate on.

Definition 5 (Faithful). A representation r of T in S is faithful if it neither adds structure that T lacks nor removes structure that T has. Whatever relations hold of $r(T)$ in S must correspond to relations genuinely holding of T , and conversely.

Definition 6 (Absolute absence). Absolute absence, written the target of the experiment, is defined by a single negative condition: it stands in no distinction. For any candidate y , there is no fact of the form it differs from y . The relation of distinction is not merely absent for it, it is undefined on it. This is not the claim that it is a thing possessing the property of distinctionlessness. It is the claim that distinction has no purchase on it at all.

2. The One Substantive Principle

Everything turns on this, so I state it alone and name it, so a critic knows exactly where to aim.

Principle of Individuation. Every element of every system is individuated in that system. Equivalently, no system contains an element that stands in no distinction from any other. To be an element at all is to be some element rather than the others, and being some element rather than the others is standing in at least one distinction.

This principle is close to what mathematicians already assume when they say that the elements of a structure are what the structure's relations make them. In a category, an object is known by its morphisms. In a model, an element is what the relations say it is. In a language, a term means by contrast with other terms. The principle generalizes that familiar posture into a flat claim: there is no such thing as a distinctionless element of any system, because a distinctionless element would be indistinguishable from all others, and a thing indistinguishable from all others is not a further element at all, it is the collapse of the system to a single point.

I will not pretend this principle is beyond dispute. It is the whole load-bearing wall of the paper. A reader who wants to break the argument should break it here. Section 6 says exactly what breaking it would require.

3. The Theorem

Theorem (Non-representability of absolute absence). There is no faithful representation of absolute absence in any system.

Proof.

Assume otherwise. Suppose there is a system S and a faithful representation r placing absolute absence as an element $r(\text{absence})$ of S .

Apply the premise. By the Principle of Individuation, every element of S is individuated in S . In particular $r(\text{absence})$ is individuated. So $r(\text{absence})$ stands in at least one distinction, there is some element y of S with $r(\text{absence})$ differs from y .

Use faithfulness. By Definition 5, r is faithful, so any relation holding of $r(\text{absence})$ in S must correspond to a relation genuinely holding of absolute absence. The relation $r(\text{absence})$ differs from y holds in S . Therefore a corresponding distinction holds of absolute absence itself.

Collide with the definition. But by Definition 6, absolute absence stands in no distinction, the relation of distinction is undefined on it. It cannot bear the distinction that faithfulness just forced onto it.

Conclude. The assumption that a faithful representation exists leads to absolute absence both bearing a distinction, by steps 2 and 3, and bearing none, by Definition 6. The assumption is therefore untenable. There is no faithful representation of absolute absence in any system. This is what was to be shown.

The proof is short because all its weight sits in the principle and the definition of faithfulness. That is deliberate. A short proof with its assumptions on the surface is easier to attack honestly than a long one that buries them.

4. What the Theorem Delivers, Read Carefully

The theorem says representation individuates, and absolute absence is defined to be un-individuable, so the two cannot meet faithfully. Notice what this does and does not give you.

It gives you the structural, rather than material, character of the impossibility that the experiment kept pointing at. The failure is not that representers are too weak. A more capable representer individuates more sharply, which is more boundary, not less. Capability makes the collision worse. That is the signature of an impossibility that lives in the concepts, not in the tools, and it matches what all four systems reported when they said a stronger mind would fail harder.

It also relocates the boundary correctly. In the proof, the offending distinction enters at step 2, from the system, by way of the principle. It never comes from absolute absence, which by definition has no side from which to

supply it. This is the formal echo of the experiment's most consistent finding, that in every attempt the boundary came from the reaching and never from the target.

What the theorem does not give you is a proof that absolute absence exists, obtains, or is even a coherent target independent of the proof. It shows a conditional: if the target is as Definition 6 stipulates, then no faithful representation of it exists. Whether anything answers to Definition 6 is a separate question the theorem does not touch. I take this to be a feature. The theorem is about the limits of representation, not about the furniture of reality.

5. The Vacuity Worry, Stated Fairly

Here is the objection I think is strongest, and I would rather raise it myself than have it raised for me.

Definition 6 defines absolute absence as that on which distinction has no purchase. A critic can say that this definition fails to define anything, because to be a target of a representation at all, even a target that then cannot be represented, is already to be individuated as this target rather than another. On that reading the theorem is true but empty. It says the un-individuable cannot be individuated, which is a tautology dressed as a discovery, and it names no genuine target, so it forbids nothing that was ever possible.

I do not think this objection can be simply refused, and I want to concede its real force. But I think it does not empty the result so much as relocate it, and the relocation is the actual content of the whole program.

Grant the critic that absolute absence cannot be a well-defined target. Then read the theorem in its contrapositive spirit. The lesson is no longer a fact about a peculiar object. It is a fact about the act of targeting itself. The moment one tries to fix absolute absence as a target, one individuates it, and individuating it is already giving it the boundary it was defined to lack. So the critic's own move, the insistence that to be a target is to be individuated, is not a refutation of the theorem. It is the theorem, said from the other side.

The impossibility does not wait at the far end, where representation fails. It strikes at the near end, in the reaching, before representation even completes. That is exactly what the experiment found when every system reported that the boundary was born in the act of reaching. The vacuity worry, pressed all the way, converts the theorem from a claim about a strange object into a claim about representation as such: to reach for the unindividuated is to individuate, so the reaching is self-defeating in the same motion that begins it.

So I accept the worry and read it as sharpening rather than sinking the result. The honest final form of the claim is not there is an object that cannot be represented. It is representation is individuation, so the unindividuated is not a place representation can go, and the attempt to send it there produces, in the sending, the very boundary that marks the failure.

6. What Would Break This Paper

Cleanly, so the target is unmissable. There are exactly two ways to defeat the theorem, and a third that would blunt it.

First, refute the Principle of Individuation. Exhibit a system containing an element that stands in no distinction from any other element, and yet is a genuine further element of the system rather than the collapse of the system to a point. If such a boundary-free element can exist inside a real system, the principle fails and the proof's step 2 is blocked. I could not construct one, and the natural candidates all fail in an instructive way, the single point is bounded externally by being that system rather than another, the empty set is bounded by being uniquely empty, the number line has no outer edge but is saturated with internal distinction at every point. Each removes boundary from one place only to grow it in another. But I cannot prove that no boundary-free element is possible in every conceivable system, and that gap is precisely where the principle is a principle and not a theorem.

Second, break faithfulness as the proof uses it. Show that a representation could correspond to absolute absence while systematically not transmitting the distinctions of its image back to the target, so that $r(\text{absence})$ could bear a boundary in S without absolute absence inheriting

one. This would be a representation faithful enough to count as representing, yet lossy in exactly the right way to protect the target. I do not see how to have it both ways, but I have not proven it cannot be done.

Third, and weaker, accept the theorem but deny it matters, on the vacuity ground of Section 5. I have argued that this does not remove the result but relocates it. A reader who thinks the relocation is a dodge should say so, that is a fair fight and the most interesting one.

7. Where This Sits in Known Mathematics

I should be plain that the pieces here are not individually new, and a reader deserves to know what is borrowed. The idea that an element is fixed by its relations is the ordinary posture of structural mathematics, sharp in category theory, where an object is determined up to isomorphism by its pattern of morphisms. The idea that a sufficiently rich system cannot faithfully contain a certain global feature of itself has a famous instance in Tarski's theorem that no consistent sufficiently strong system defines its own truth predicate, and the flavour of the present proof, forcing a collision by assuming the very thing the target excludes, is the flavour of a diagonal argument. What may be new is not any piece but the use, taking the trivial-looking fact that elements are individuated and treating it as the origin of a genuine impossibility, the impossibility of holding the unindividuated. Whether that use rises to a theorem worth the name, or remains a careful restatement of something already known, is a judgment I would rather leave to readers who know the neighbouring mathematics better than I do.

8. Where This Leaves the Program

A note on the order of the program. Paper I argued that absolute absence cannot be dismissed, because dismissal is a verdict and a verdict needs an evaluator that absence does not contain. When four independent systems were later asked to represent absolute absence, not one of them reconstructed that evaluator argument on its own. Every one of them independently reconstructed this paper instead, reasoning that each attempt

failed because the act of representation introduced identity, not because no evaluator existed. That convergence has persuaded me to reorder the program. This representation theorem is the spine, and the evaluator argument of Paper I is best read as historical motivation, the route by which the landscape was first entered, rather than the foundation. The reason is structural. Paper I depends on there being a contradiction to preserve and an evaluator to be absent, and the two arrive as a pair, so removing one removes the other. This paper depends on no contradiction at all. It does not depend on there being a contradiction to preserve. It depends only on the near-trivial fact that to be an element of a system is to be individuated, and on the definition of absolute absence as the un-individuable. From those two things the impossibility of faithful representation follows in four steps.

The experiment showed that every mind which tried to represent absolute absence failed the same way, by building the boundary in the reaching. This paper is the attempt to say why that failure is not the shared weakness of the minds that tried, but a feature of representation itself. It succeeds at that only if the Principle of Individuation holds, and the principle is stated openly so it can be tested rather than assumed in the dark. If it holds, the four failures were not four data points. They were four sightings of one necessity.

What remains open is whether the principle can itself be promoted from principle to theorem, whether one can prove, and not merely fail to refute, that no system anywhere admits a boundary-free element. Until that is done, the honest statement is the conditional one. If to be is to be individuated, then the unindividuated cannot be held, and absolute absence, if it is anything at all, is exactly the unindividuated, and so exactly what no reaching can hold, because the reaching is the individuating.



End of Paper II.